Spatial Coherence

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Virtual Human Interaction Lab
Symbolic Systems Honors Thesis Presentation
June 1st, 2026

VIRTUAL HUMAN INTERACTION LAB

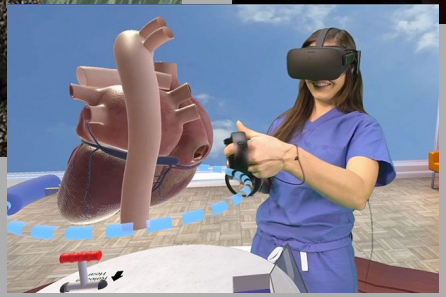
VR
Virtual Reality

MR
Mixed Reality











Three Big (Unanswerable) Questions

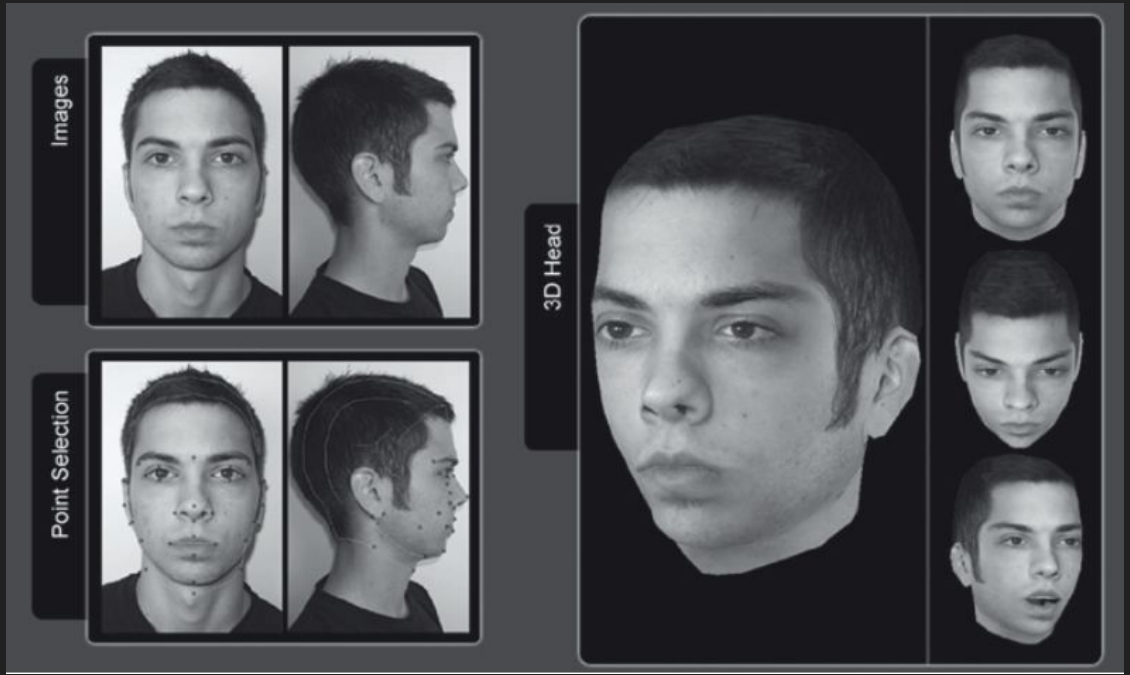
1. What if avatars could look and act photorealistically?






Amy
Sara




Images
Point Selection
3D Head

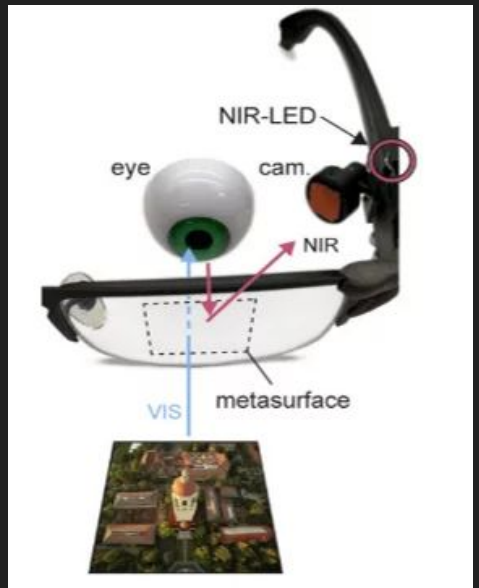



A
B

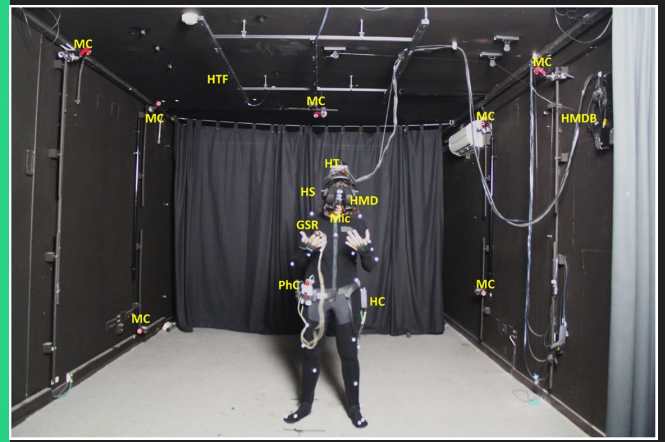







NIR-LED
eye
cam
NIR
VIS
metasurface




MC
HTF
MC
MC
MC
MC
HMDB
HT
HS
HMD
GSR
MC
PhC
HC
MC
MC

2. How does *spatiality itself* impact conversation?

◀ Outlook 9:27 AM Thu Apr 8

BC Calculus 20-21

6. The Maclaurin series for a function f is given by

converges to $f(x)$ for $|x| < R$, where R is the ra

(a) Use the ratio test to find R.

(b) Write the first four nonzero terms of the Mac
rational function for $|x| < R$.

(c) Write the first four nonzero terms of the Mac
the third-degree Taylor polynomial for $g(x)$






Workrooms Meeting
Sharing screen
10:05 AM



3. How does MR differ from VR?




PASSTHROUGH: Main Quad

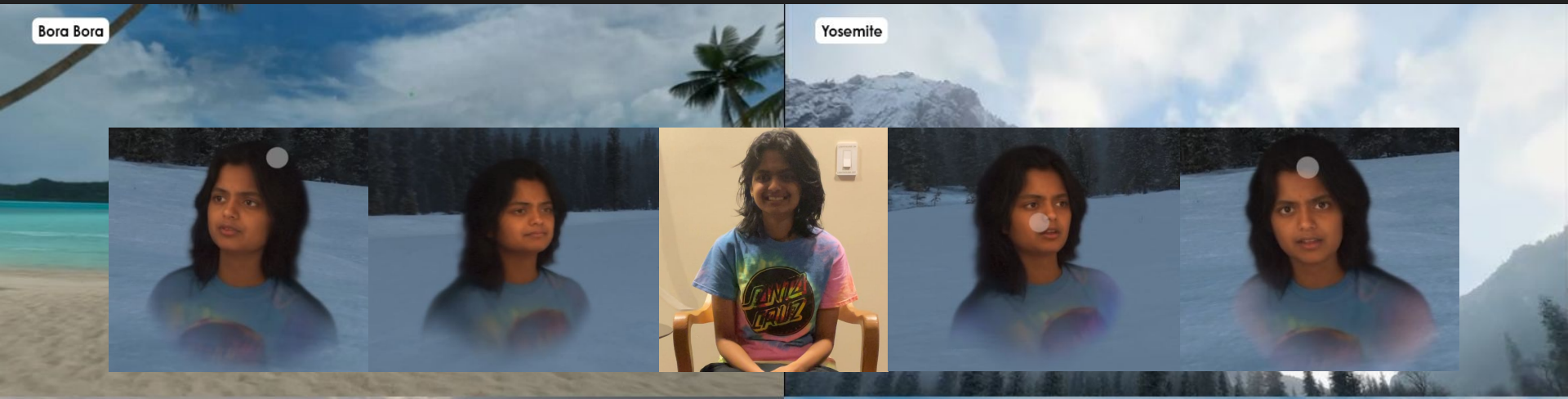

STANFORD MAIN QUAD ARCHIVES
EST. 1891
FOUNDERS' ARCADE
VIRTUAL RECONSTRUCTION

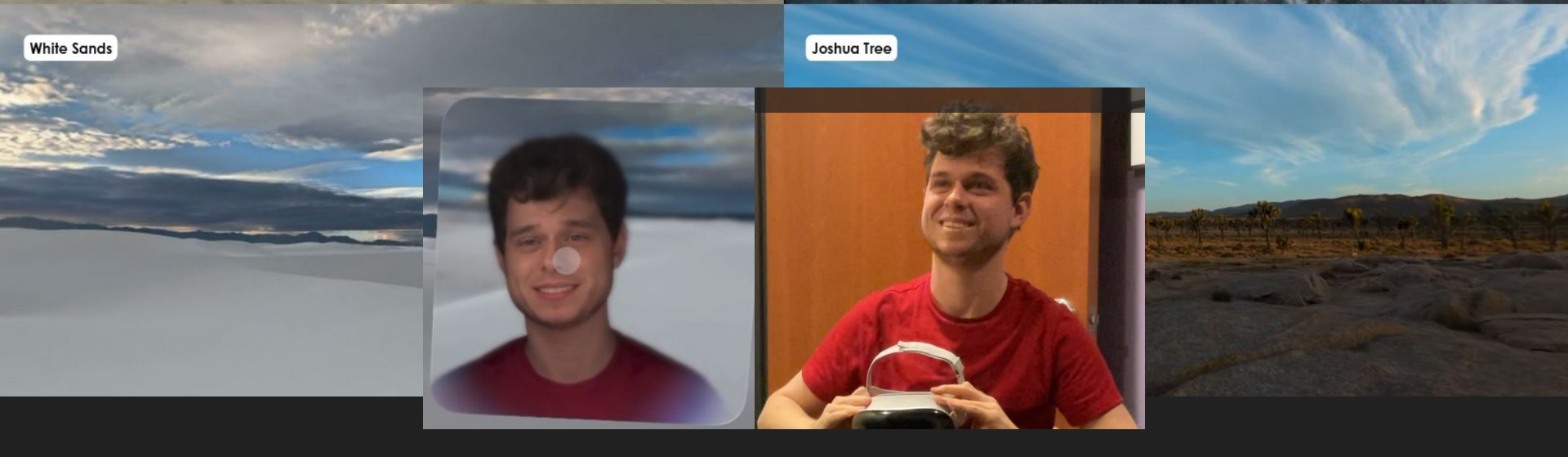
Three Big (Unanswerable) Questions

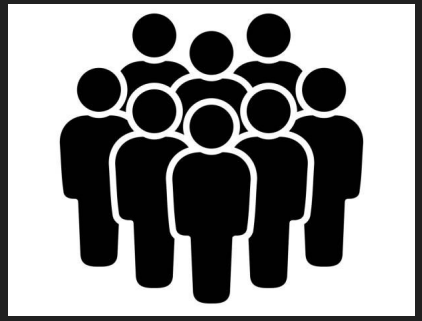
A rigorous group study with…

1. Photorealistic, fully tracked, high resemblance avatars
2. A way to isolate spatiality with no confounds
3. Comparable fidelity across MR,VR

Three Big (Unanswerable) Questions


Bora Bora
Yosemite


White Sands
Joshua Tree



n = 143, 46 groups varying familiarity

over 160 hours of video data!

1. ⭐ Walk in the door
2. Informed consent
3. Demographics & familiarity survey
4. Photorealistic avatar creation
5. Gaze calibration & interaction training
6. Enter separate, commonly configured rooms
7. POV and external body movement recording
8. Join a group call using the avatars
9. Call configuration based on group condition
10. Icebreaker activity
11. 3 minutes unmoderated conversation
12. Consensus-building conversational task explanation
13. 13 minutes for task completion
14. ⭐ Final questionnaire
15. Compensation

Analyzed Data

- **Questionnaires**
  - Multiple choice & free-response
  - Measures **co-presence, entitativity, ostracism, closeness, dis/comfort**
  - **Results**
    - Co-presence: No significant difference for IVs!
    - Physical & cognitive load: No significant difference for IVs!
    - Smaller group → felt more like a team
    - Larger group → higher ostracism

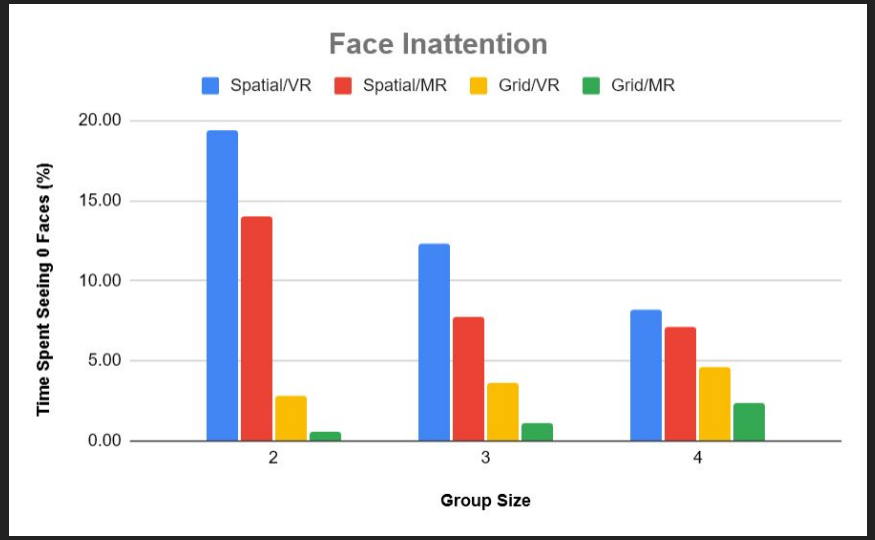
- **Face attention**
  - Point-of-view recorded → computer vision pipeline to detect visible (avatar) faces
  - Measures **patterns of attention to group members**
  - **Results**
    - Spatial (contra Grid)
      - Saw fewer faces; less time seeing whole group; **more time seeing nobody**
    - VR (contra MR)
      - Saw fewer faces overall
      - *Spent more time seeing nobody*

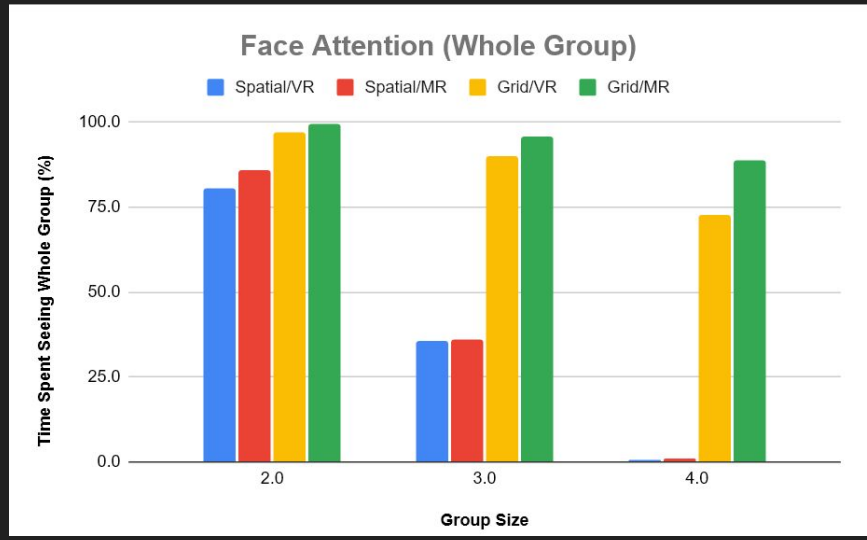
Data To-Be-Analyzed

- Gaze data
  - Visible gaze/focal subject marker in recorded POV
  - Developed schema for manual coding
  - Measures **precise attentional patterns & eye movement**
  - **“What are people looking at if not faces?”**
- Conversational data
  - Speech recorded, manual transcription
  - Measures **conversation quality, group dynamics, & more**
  - **“How does the way people talk about themselves & the group differ?”**
- Body motion data
  - Body movements recorded → Computer vision analysis
  - Measures **synchrony** between group members
  - **“Are people more in sync with remote collaborators in x or y condition?”**

Conclusions:

- Spatiality and VR both lead to **paying less attention to others** but don't impact self-reported group dynamics
- Modality is less important for consensus tasks than how many people are involved
- …and more to come!





Thank You!

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